

27 September 2026

Wellcome Leap
Resistance Networks Program

Re: Support for “Universal properties of plasmid evolution and robust risk forecasting for preemptive pARG intervention”

Dear Members of the Review Committee:

I am writing in support of Dr. Philip J. Gerrish’s proposal to the Wellcome Leap Resistance Networks program.

I served on Phil’s doctoral thesis committee in Richard Lenski’s laboratory, and we have collaborated for nearly three decades. Our work began with the experimental study of evolving mutation rates in *E. coli*, published in *Nature*. We subsequently worked together on how genetic linkage affects natural selection, the fate of beneficial mutations in asexual populations, and the possibility of forecasting evolutionary change. This sustained collaboration gives me a substantial basis for assessing Phil’s work.

A recurring strength of Phil’s research is his ability to identify a consequential evolutionary mechanism, formulate it mathematically, and determine what evidence would test it. His proposal brings that approach to resistance plasmids. The question of how selection on a bacterial host affects a linked plasmid, and how horizontal transfer changes that relationship, is one he is particularly well prepared to address.

The project is ambitious. Its value will depend on testing the theory against observed plasmid dynamics and establishing whether it improves prediction. Phil understands that requirement, and his record gives me confidence in his ability to pursue it with rigor. I strongly support his proposal.

Sincerely,



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